Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Course: SWE-301: Agile Software Engineering
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Project: Coffee-Kiosk (CK Board)
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Part 1: Epics & User Stories Overview

Epic 1: Customer Ordering Experience (CK-E1)

Description: Enable customers to browse, customize, and place coffee orders through an intuitive self-service interface.

Key	Story	Pts	Priority	User Story
CK-1	Coffee Selection	3	High	As a customer, I want to browse and select from different coffee types, so that I can choose my preferred beverage.
CK-2	Drink Size Selection	2	Medium	As a customer, I want to choose my drink size, so that I can get the right portion for my needs.
CK-3	Customization & Milk Add-ons	5	High	As a customer with dietary preferences, I want to customize milk choices and syrup flavors, so that I receive my customized drink accurately.
CK-4	Order Review & Cart Summary	3	High	As a customer, I want to review my selected items, customizations, and pricing before paying, so that I can confirm my order without errors.

Epic 2: Payment & Checkout Processing (CK-E2)

Description: Provide secure, seamless digital payment options, contactless card reader integration, and instant receipt generation.

Key	Story	Pts	Priority	User Story
CK-5	Contactless & Credit Card Payment Gateway	5	High	As a customer, I want to pay using contactless credit card or Apple Pay/Google Pay, so that I can complete payment quickly and securely.
CK-6	Digital Receipt & Order Number	2	Medium	As a customer, I want to receive an on-screen order number and SMS digital receipt, so that I have proof of purchase and can claim my drink.

Epic 3: Barista Order Fulfillment & Queue (CK-E3)

Description: Provide real-time order queue visibility for baristas and pickup status notifications for customers.

Key	Story	Pts	Priority	User Story
CK-7	Real-time Barista Kitchen Display Queue	5	Medium	Sprint 2 Backlog
CK-8	Customer Order Ready Display Board	3	Low	Sprint 2 Backlog

Part 2: Answers to Reflection Questions

1. Did your estimations reflect the actual effort?

Yes, the team's story point estimations using the Fibonacci sequence (1, 2, 3, 5, 8, 13) accurately reflected the actual relative effort, complexity, and technical uncertainty encountered during Sprint 1.

Specifically, assigning 5 story points to CK-3 (Milk and Syrup Customization) and CK-5 (Contactless Payment Gateway) was fully justified. CK-5 required hardware SDK interfacing with contactless card readers, handling timeout protocols, and encrypted PCI compliance tokens, which represented the highest technical complexity in the sprint. CK-3 involved dynamic client-side pricing calculation and dietary modifier rules. In contrast, simpler stories such as CK-2 (Size Selection, 2 pts) and CK-6 (Digital Receipt, 2 pts) had straightforward UI requirements and were completed swiftly without architectural bottlenecks. Planning Poker prevented individual estimation bias and allowed the team to align on risk factors upfront.

2. Was your backlog well-prioritized?

Yes, the backlog was systematically prioritized using a customer-journey and business-value framework (MoSCoW criteria).

Sprint 1 intentionally committed the end-to-end Minimum Viable Product (MVP) ordering loop: Menu Selection (CK-1) → Sizing (CK-2) → Customization (CK-3) → Order Review (CK-4) → Payment (CK-5) → Receipt (CK-6). Delivering this closed customer transaction flow first was critical because a self-service kiosk has zero commercial utility if a customer cannot select an item and complete payment.

Secondary operational requirements, including Epic 3's Kitchen Display Monitor (CK-7) and Customer Pickup Status Board (CK-8), were prioritized for Sprint 2. This strict prioritization avoided premature optimization of backend queue screens before the primary customer intake mechanism was validated.

3. How did your simulated sprint align with your plan?

The simulated sprint closely aligned with our sprint planning forecast, achieving a 100% completion rate (20 of 20 planned story points delivered). The sprint timeline progressed as follows:

- **Day 1:** Sprint kickoff with all 6 stories committed in 'To Do'. Work initiated on high-complexity items (CK-3 and CK-5).
- **Day 2-3:** Early completion of CK-3 (5 pts) and CK-1 (3 pts), dropping remaining work to 12 pts and creating a critical buffer.
- **Day 4:** Contactless payment gateway (CK-5, 5 pts) successfully integrated and validated on sandbox hardware.
- **Day 5-6:** UI review cart (CK-4, 3 pts) and sizing controls (CK-2, 2 pts) verified.
- **Day 7:** Digital receipt generator (CK-6, 2 pts) finalized, successfully burning the sprint down to 0 remaining points.

The team maintained a disciplined WIP (Work In Progress) limit of no more than 2-3 concurrent stories, preventing multitasking bottlenecks and ensuring items moved steadily across the Scrum board into 'Done'.

4. What insights did the burndown chart give about your team's capacity?

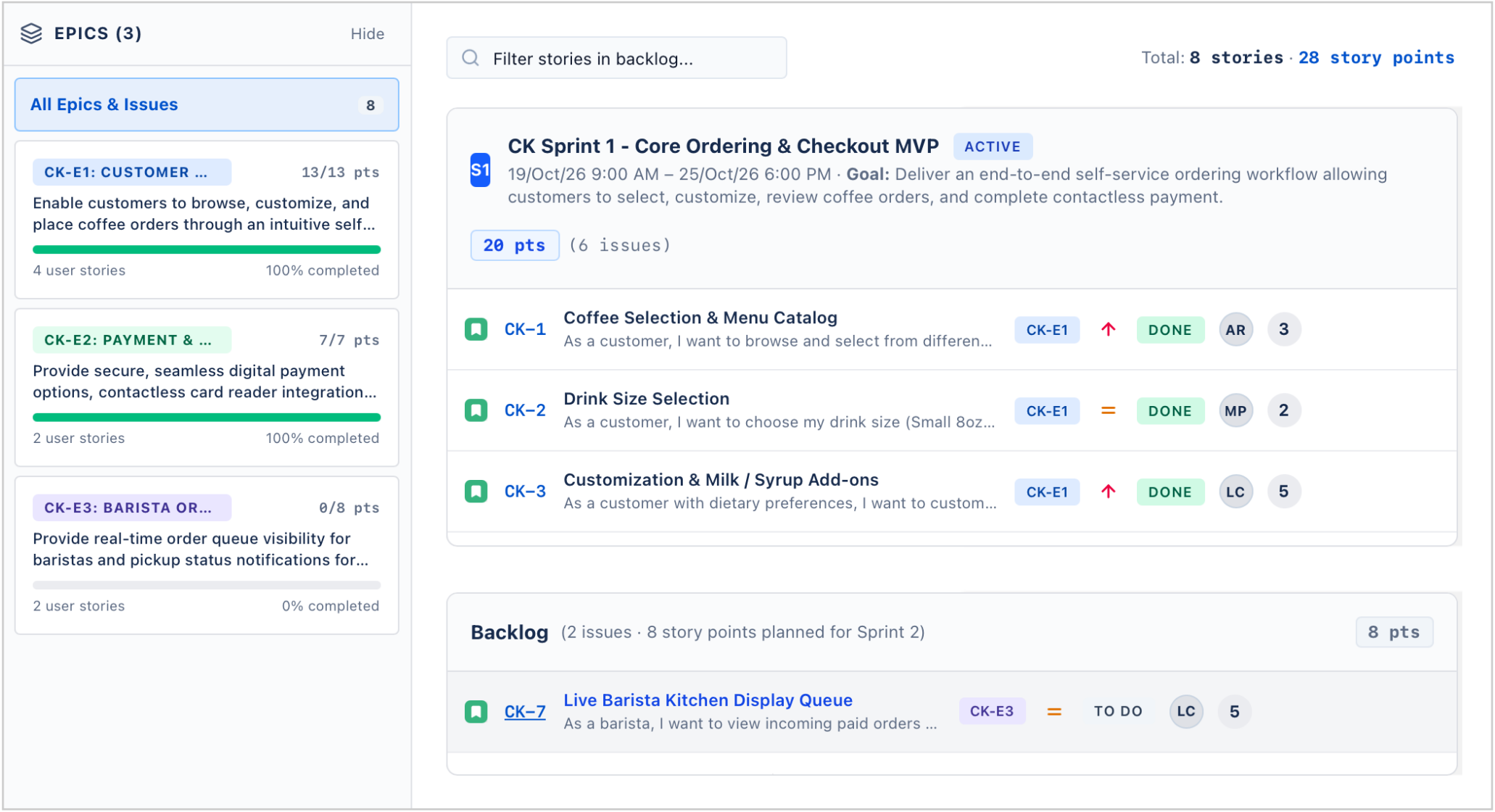
The Jira Burndown Chart provided three critical operational insights:

- **a) Established Team Velocity:** The team successfully demonstrated a stable velocity of 20 story points per 1-week sprint across cross-functional frontend, backend, and UI roles. This gives high

predictability for forecasting future sprint commitments.

- **b) Stair-Step vs Ideal Linear Trajectory:** The ideal guideline indicated an expected burn rate of 2.86 points per day. Because user stories represent indivisible units of value that only burn points upon reaching 'Done', the actual burndown exhibited a realistic stair-step descent. The team remained consistently on or below the guideline line, proving that work was neither overloaded nor stalling.
- **c) Zero Scope Creep & Sprint Discipline:** The remaining points line did not experience any upward spikes, confirming that the Product Owner and Scrum Master successfully protected the sprint backlog from unplanned scope creep during active execution.
- **d) Capacity for Sprint 2:** With the remaining backlog standing at 8 story points (CK-7 and CK-8), the team has ample capacity in Sprint 2 to complete Epic 3 and address refinement or end-to-end hardware stress testing.

Deliverable 1: Jira Backlog - Epics & User Stories



Deliverable 2: Story Point Assignments (Fibonacci)

Total Backlog Points

28 pts

Across 8 user stories

Sprint 1 Commitment

20 pts

6 stories committed (MVP Scope)

Sprint 2 Backlog Queue

8 pts

2 stories (Kitchen queue & display)

Estimation Scale

Fibonacci Series

1, 2, 3, 5, 8, 13 (Non-linear)

Fibonacci Estimation Scale Reference (Relative Sizing)

Triad Criteria: **Effort + Complexity + Uncertainty**

1 TRIVIAL
Negligible risk, minimal code changes, zero unknowns.

2 SMALL
Simple UI components, well-documented (e.g. Size Selection).

3 MEDIUM
Standard business logic with validation (e.g. Menu catalog, Order review).

5 COMPLEX
Multiple dependencies, integrations, or edge cases (e.g. Payment, Customizer).

8 HIGH UNCERTAINTY
Significant unknowns, cross-team blockers. Split recommended.

13 EPIC / TOO LARGE
Massive body of work; must be split into multiple user stories.

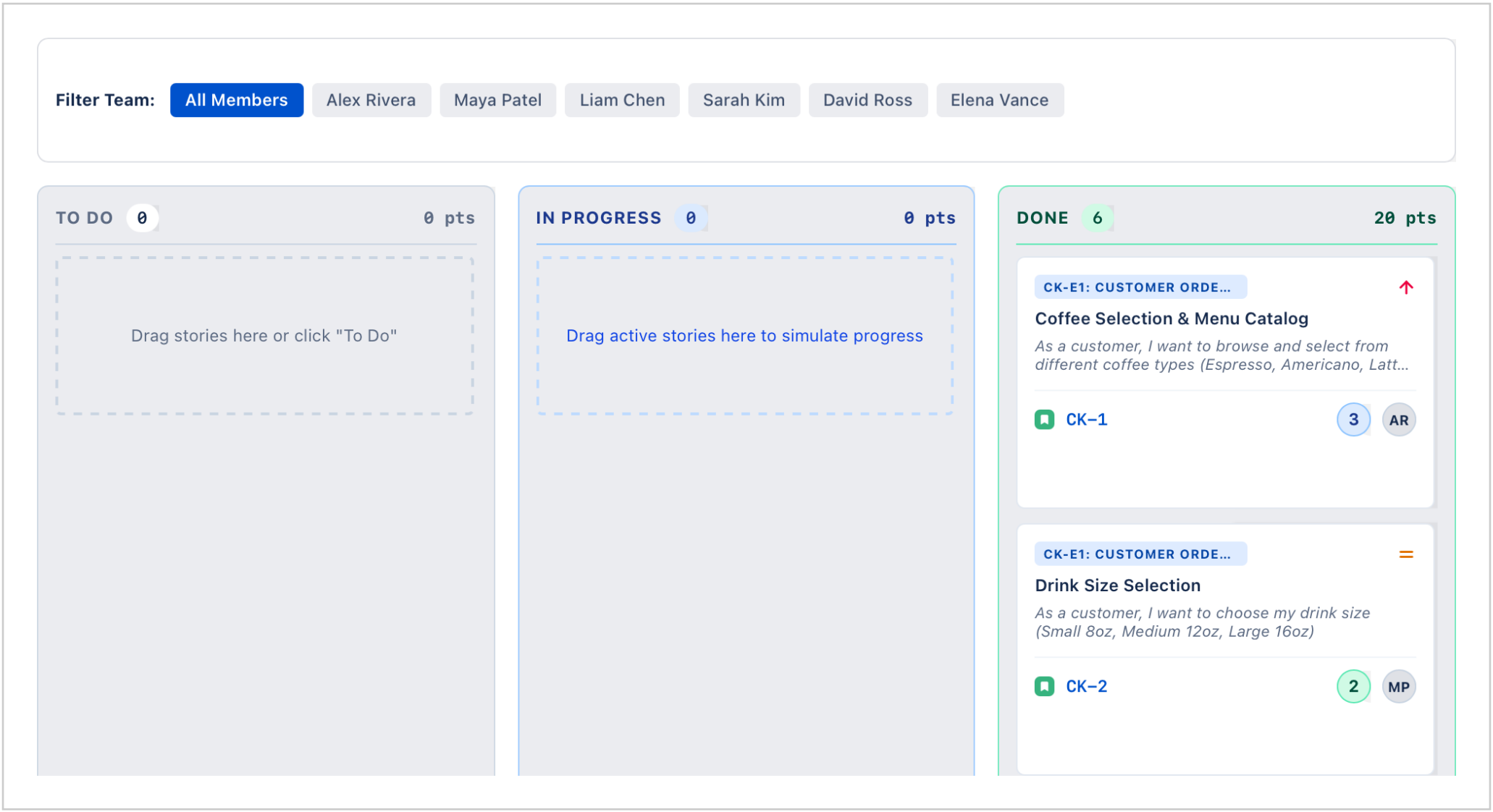
Comprehensive Story Point Assignment Matrix

Every user story estimated against complexity, implementation effort, and risk using Planning Poker consensus.

Lab 2 Software Requirement Specification

Key	User Story & Summary	Epic Theme	Complexity	Effort	Uncertainty	Story Points	Sprint Phase
CK–1	**Coffee Selection & Menu Catalog** *As a customer, I want to browse and select from different coffee types*	CK-E1: Customer Or...	Moderate	Medium	Low	3	Sprint 1 (Active)

Deliverable 3: Active Sprint Board (Scrum)



Deliverable 4: Sprint Burndown Chart

SPRINT:

CK Sprint 1 - Core Ordering & Checkout MVP Completed

ESTIMATION STATISTIC:

Story Points

Sprint Duration:
19/Oct/26 9:00 AM – 25/Oct/26 6:00 PM

Committed Points	Completed Points	Planned Daily Burn	Final Velocity
20 pts	20 pts (100%)	2.86 pts/day	20 pts / sprint
Sprint backlog forecast	All 6 stories delivered	Ideal linear burn rate	Zero scope carryover

Sprint Burndown Progression

Remaining Story Points vs Ideal Guideline over 7-day Sprint timebox

